

ECONOMICS 490 UNDERGRADUATE THESIS

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Abstract

The central theme of this paper is the use of statistical analysis in the realm of sports, particularly football. Football has become a sport filled with mathematical intricacies delving into the more quantitative aspects of what makes a good player valuable. The purpose of this paper is to answer the question would Lionel Messi still be considered one of the most valuable players in the world if he were to play for the English Premier League instead of La Liga. To help answer this question a three step model was devised, including the use of two found regressions and an already established mathematical model. The first step was a Poisson Regression model to estimate the number of goals of each match during a given season, in this case 2016 to 2017 of La Liga, with and without Messi. Once all the estimated goals per match were found for both data sets, the ELO ratings of every team in the season were calculated, one with Messi and one without. Both ELO ratings of Messi's team, one with Messi and one without, were taken and converted to adjusted La Liga ELO ratings against Premier League teams, illustrating not only how Messi's team stands against other Premier League teams, but also how they fair if he was not present in order to help determine his overall impact. The results were found to be that with Messi playing the ELO rating was 1539.57, ranking better than all other Premier League teams except for one, and without him, the ELO rating of Barcelona was found to be 1506.74 ranking far lower and just near the threshold of being considered an average team. The purpose of this paper is to not only establish a background on the topic of discussion, but also how the conclusion has been reached.

Dedication

This paper is dedicated to both the Vancouver School of Economics and the department of Computer Science for giving me the tools needed to succeed in this world.

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Chapter 1

Introduction

Football is considered to be one of the most popular and widely watched sports in the world. Garnering such a massive fan base that ranges from continent to continent, the sport has touched the lives of many and through it's spread of entertainment has come a level of statistical and mathematical sophistication. As the role of data analysis becomes more prominent in a number of industries, this tool has proven to be beneficial in the realm of sports leading to a recently created sub-branch of data analytics more commonly known as sports analytics. Such a mechanism has become a favoured technique in analyzing a player's performance, a team's ability to win, and determining the value of a player in a hypothetical surrounding. In football, many define a top player as one that has not only amassed a wide variety of success within the sport, but also to be considered an immeasurable asset that cannot be matched by any other player. Any such player that has been considered to meet this definition by the majority, has also come under an immense amount of scrutiny leading many to debate, analyse, and critique such a player's ability to be considered a top notch player.

Lionel Messi is an Argentinian football player who since the age of sixteen has done nothing but surpass expectations and break records. Throughout his career, many have labeled him as one of the top players in football. However, with such levels of praise have also come high levels of criticism. The Argentinian has always

played for one football club, Barcelona F.C., a Spanish club that participates in the annual La Liga or Spanish League. La Liga has been a league primarily dominated by two teams, Barcelona F.C. and Real Madrid, with one or the other being the most likely to win the league during a season. On the other hand, the Premier League or English League has been considered to be one of the most unpredictable leagues in football due to the competitiveness and wealth of many of the teams that compete. Teams that were once considered top tier one season, could be nearly relegated next due to a series of factors absent in many other leagues. In order to determine the validity of Lionel Messi's ability, a simulation of him playing against Premier League teams has been conducted in order to help determine whether or not he would still be considered a valuable player. To formalize the focus of this paper in more structured terms the question that will be answered is would Lionel Messi still be considered one of the most valuable players in the world if he were to play for the English Premier League instead of the Spanish league.

In order to help answer this question a simulation was created that made use of the interaction of three separate models. The first model was the Poisson model, a model designed to determine the expected number of goals of every match in a season based on Lionel Messi's presence and the aggressive nature of the match. This model was utilized to not only predict the expected goals of all matches in a season where Messi was playing, but also to predict what the results would be if Messi was removed from that very same reason. This resulted in the creation of two datasets, one which had the expected goals of every match in a season with Messi present and the other without Messi being present. Using these expected goals from both datasets, ELO ratings were calculated for both sets respectively, showcasing the impact of Messi on the rating. ELO Ratings are an existing mathematical model that helps rank teams during a season based on their performance throughout an entire season. Once these two ELO Ratings datasets were created for every team, the ELO ratings of Barcelona, were then converted to La Liga ratings adjusted for

the Premier League using a simple linear regression model. The technical challenges that were faced were simply determining how all three of these models would work with one another in such a way that it would create a coherent simulation. The model came to the conclusion that Barcelona's adjusted ELO Rating with Messi is 1539.57 and without him is 1506.74. The results found do align with the initial intuition as even though the Spanish League is considered a far less aggressive league, it does not necessarily mean it is an easier league and thus while Messi's ELO rating is not the highest when compared towards teams in the Premier League, it is far higher than many other Premier League teams. Additionally his effectiveness is also demonstrated by the fact that his team's adjusted ELO rating without him is far lower than if he was present. Both of these thus convey that he is not only crucial to his team but also an invaluable player.

To delve into how these models interact and how the conclusion was attained in a more detailed setting, this paper will first establish the general consensus around the topic of study. Secondly, it will dive into the intricacies of each model, how the model works individually, the variables of the model, and how the models complement one another in order to implement a simulation of Messi playing against Premier League teams through the use of another league known as the Champions League. For the purpose of being concise, the English Premier League will always be referred to as the EPL and the Spanish League will be referred to as La Liga.

Chapter 2

Background

The emphasis on this paper is to determine the value of a player if he were to play in another competition. Every country in Europe has an internal football league played by 20 teams, each representing a city within the country. A league consists of seasons, where a season takes place throughout the year with each team playing against another team twice, thus resulting in a total of 380 matches in a year. A player's value is determined by his effectiveness and contributions in the team he currently plays in. As Lionel Messi is a striker, a crucial position on the team responsible for scoring goals, a player's value for the context of this paper will be defined as the number of goals his team is able to score while he is present.

Even though there's a lack of formal research for this particular question. Before delving into why Messi would struggle, certain factors of both La Liga and the EPL need to be highlighted in order to delineate the differences that cause the latter to be more competitive than the former. An article from the guardian analyses certain quantitative factors as to why two teams dominate such a vast league. A valid feature that is utilized to bolster this claim are the salary caps, while teams such as 'Barcelona and Real Madrid have budgets of around 632.9 million and 566.5 million euros respectively, the third wealthiest team, Valencia, trails by a near 300 million', thus inhibiting many of these teams to retain valuable individuals as they simply transfer to teams willing to meet their salary quotas (Lowe). To further

reinforce this point, an ex-manager of Valencia has claimed that “It would be lovely if everything became more equal,” referring to the substantial division between La Liga club’s wealth (Lowe). A number of these points thus provides a firm basis as to why the EPL is considered a more unpredictable league compared towards its counterpart La Liga.

While the quantitative side of Lionel Messi’s ability is being measured in this paper. A qualitative outlook does give an insight as to why many consider his success as being partially attributed towards the competition he plays in. Firstly, an article cites that Cristiano Ronaldo, a player who made the transition from the English Premier League to La Liga, believed that “Messi’s woes for the Argentina national team is proof he can only perform at his best in the comfortable confines of the Catalan club” (Burke). However another side to the argument favours Messi’s abilities. An article by bbc favours the side that Messi could perform in the Premier League stating that “the 31-year-old has scored 22 goals and delivered six assists in 28 games against England’s finest” (BBC). This showcases that even Messi yet performs even against English teams. Additionally, other players have also voiced their opinion on the matter at hand, Spanish player Xavier Hernandez has stated “I think he’d be even better in the Premier League because it is not so defensive.” Another claim reinforcing this other side of the argument. Lastly, Lionel Messi himself has “revealed that he would do well in the Premier League” (John Owen Nwachukwu). In short, one can thus witness that claims on both sides have been made, but due to their more qualitative aspects a certain level of bias makes the debate a more word against word scenario. Nonetheless, the objective of this paper is to help reinforce a side through quantitative results free from subjectivity.

Chapter 3

Data

In order to come towards a formal conclusion, a number of data sets were analyzed to narrow down relevant features. The primary data sets that were made use of were La Liga, EPL, and Champions League season statistics. These data sets consisted of every match played during a given season, consisting of several variables, such as the number of goals scored by each team, the result of the match, the number of fouls, red cards, and yellow cards acquired during a game. For the purpose of designing the model initially, three core data sets were utilized. The first was for the La Liga season of 2016 to 2017. The second was for the EPL season of 2016 to 2017. And the third data set consists of all games played between the year 2010 and 2017 for the Champions League, a tournament played by teams from different leagues.

The first data set and second data set sketches all the matches that were played during the season of 2016 to 2017 for both the La Liga and EPL respectively. Both data sets were found on datahub.io. The first data set was used to help implement a Poisson Regression Model and to help design a script to calculate ELO ratings. Whereas the second was to help draft basic statistical comparisons between both leagues. The reason for the emphasis on one season was the simplicity of which it allowed to design a concrete model, which at the same time was extendable towards the addition of more and more seasons. While each data set contains 19 different

features, the primary ones of focus were selected to be the number of goals, the number of fouls committed, the number of yellow cards, and the number red cards, each split into columns of two, one for the home team and the other for the away team. A sample of the data set can be seen in figure 3.1. where each row is a match and each column represents the type of data that was selected as mentioned before. All features containing an H correspond to the home team and all features containing an A correspond with the away team. Each row is a single match and the data is ordered from earliest match played in the season to the latest. FTHG and FTAG are the number of goals scored after the match was completed by each team. HF and AF are the number of fouls by each team during the match. HY and AY are the number of yellow cards attained by each team during the match. HR and AR are the number of red cards garnered by each team during the match.

Date	HomeTeam	AwayTeam	FTHG	FTAG	HTR	FTR	AF	HY	AY	HR	AR
8/19/16	La Coruna	Eibar	2	1	D	H	17	3	3	0	0
8/19/16	Malaga	Osasuna	1	1	D	D	22	2	3	0	0
8/20/16	Barcelona	Betis	6	2	H	H	13	1	3	0	0
8/20/16	Granada	Villarreal	1	1	D	D	11	2	3	0	0
8/20/16	Sevilla	Espanol	6	4	D	H	15	2	2	0	0
8/21/16	Ath Madrid	Alaves	1	1	D	D	15	0	4	0	0
8/21/16	Sociedad	Real Madrid	0	3	A	A	11	2	3	0	0

Figure 3.1: Sample Data Set: D1 and D2

Through these data sets mentioned above, basic statistical comparisons were made to help highlight the differences between La Liga and the EPL. Summaries of all the selected features for both La Liga and EPL can be seen in figure 3.2 and 3.3.

Variable	Obs	Mean	Std. Dev.	Min	Max
fthg	380	1.663158	1.40756	0	7
ftag	380	1.278947	1.216023	0	6
hf	380	14.06842	4.356241	1	28
af	380	13.82105	3.985406	4	26
hy	380	2.347368	1.465656	0	7
ay	380	2.65	1.416497	0	7
hr	380	.0894737	.3037058	0	2
ar	380	.1421053	.3715699	0	2

Figure 3.2: Summary Statistics for La Liga

Variable	Obs	Mean	Std. Dev.	Min	Max
fthg	380	1.597368	1.306971	0	6
ftag	380	1.202632	1.227862	0	7
hf	380	11.06316	3.468853	2	24
af	380	11.65789	3.556378	3	24
hy	380	1.744737	1.256371	0	6
ay	380	1.886842	1.301647	0	6
hr	380	.0552632	.2400499	0	2
ar	380	.0526316	.2235913	0	1

Figure 3.3: Summary Statistics for EPL

Figure 3.2 and 3.3 conveys that teams in La Liga, on average, score more goals than that of EPL when playing at home and away. Moreover, the number of fouls, yellow cards, and red cards in La Liga are, on average, much higher than that in the EPL, with average of fouls at home and away for La Liga differing by a near 3 additional fouls compared to that of fouls at home and away in the EPL. While a logical conclusion to draw is that La Liga is considered more aggressive, a more likely scenario is that La Liga is far softer. This is because even though more fouls have been committed and more yellow and red cards have been awarded the number of goals still being scored is higher in La Liga than that of the EPL, thus it can be inferred that referees in La Liga are more likely to report fouls and award penalties than those in the EPL. Because of the lack of reported fouls in the EPL, players are able to get away relatively easier thus making defenders more aggressive and disabling the ability to score more goals. On the other hand, La Liga labelling is more prominent, restricting defenders to use a more passive playing style thus enabling strikers to score more goals.

The first data set was restructured further to help design the Poisson Regression and determine the effect of Lionel Messi on the number of goals scored by his team, Barcelona. Figure 3.4 showcases a sample of the data set with these changes below. The first was that every match was counted twice, where a new home variable was introduced to determine whether or not that team was playing at home. The second time the match was counted was to showcase it in perspective of the opponent.

The Goals is the number of goals scored by the perspective team during the match. Additionally the two new added variables are to help integrate Lionel Messi into the data, the first is whether or not he played in a particular match at home and the second was whether he was playing away. The Playedaway variable was to enable his effect on both the home and away goals, as the played variable simply gives the coefficient of the number of goals he scored while he was playing at home. This data was inputted manually by going through every match to see whether or not he was present.

Date	Team	Opponent	Goals	Home	Played	Playedaway
8/19/16	La Coruna	Eibar	2	1	0	0
8/19/16	Malaga	Osasuna	1	1	0	0
8/20/16	Barcelona	Betis	6	1	1	0
8/20/16	Granada	Villarreal	1	1	0	0
8/20/16	Sevilla	Espanol	6	1	0	0
8/19/16	Eibar	La Coruna	1	0	0	0
8/19/16	Osasuna	Malaga	1	0	0	0
8/20/16	Betis	Barcelona	2	0	0	0
8/20/16	Villarreal	Granada	1	0	0	0
8/20/16	Espanol	Sevilla	4	0	0	0

Figure 3.4: Sample Data Set restructured

The last data set was a Champions League data, a League where EPL teams play against La Liga teams, thus illustrating how La Liga perform against EPL teams. This similar to the first two, recorded each match played between all teams in that tournament, This data set was used to generate a simple linear regression model to help convert ELO ratings of Barcelona with and without Messi to ELO ratings adjusted for EPL teams being taken into account. Figure 3.5 showcases a sample of this data set below. This data set was attained by manually finding all matches that occurred during the Champions League between teams in La Liga and teams in the EPL.

HomeTeam	AwayTeam	FTHG	FTAG	FTR
Manchester	Barcelona	2	2	D
Barcelona	Manchester	4	0	H
Newcastle	Barcelona	3	2	H
Barcelona	Newcastle	1	0	H
Manchester	Barcelona	3	3	D
Barcelona	Manchester	3	3	D
Barcelona	Arsenal	1	1	D

Figure 3.5: Sample Data Set Champions League

All data sets can be found in the appendix submitted separately

Chapter 4

Model

In order for a conclusion to be reached a model was designed to create a simulation to determine Messi's value in the Premier League. The model centers around three steps, with each step being explored in detail. Step one was to create a Poisson Regression model that predicts the goals of every match during a given season in this case season 2016 to 2017. This regression includes a Lionel Messi coefficient to determine his effect in the expected goals for every match. Once this regression is run on the data sets, two expected goals data sets will be outputted. One where Lionel Messi was present in the season and another where he was removed purposely to determine the expected goals without him. Step two is to take each data set and use the predicted goals to calculate ELO ratings of Messi's team Barcelona with and without him. Step three is once both of these ELO ratings are determined, they are then converted to La Liga ELO ratings adjusted for when they play against EPL teams utilizing a simple linear regression model to evaluate Messi's impact.

The first step is the Poisson Regression model. The reason for making use of a Poisson Regression is because of it's ability to estimate goals and sketch the strength of each team through training the regression on a given season, due to it being a count model, determining all this based on the number of times it's seen certain aspects. In order to implement this regression the data set, a sample illustrated

in Figure 3.1, was reassembled to Figure 3.4. The basic regression model can be seen below and was attained from <https://opisthokonta.net/?p=276>, with an example of how to set the data to fit such a regression. While the author makes use of R to create this regression, I made use of fitting the regression utilizing python to enable the addition of multiple features. The basic regression model is seen in 4.1.

$$\log_e \hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \hat{\beta}_3 X_3 + \hat{\epsilon} \quad (4.1)$$

The output or dependent variable for this regression is the logarithm of the expected number of goals, represented by $\log_e \hat{Y}$ and the independent variables are the home X_1 , team X_2 , and the opponent X_3 . The home variable is a binary number, a 1 or 0. As each match is double counted to consider the perspective of both sides, when the match is seen for the first time, in terms of the home team, this variable is a 1, and when it is seen the second time in terms of the away team the variable is a 0. The second variable is the team which is simply the perspective team in a given match. Lastly, the opponent is who they were playing against. The model was extended to include the presence of Lionel Messi, culminating in the core model made use for this paper. This can be seen in the equation 4.2.

$$\log_e \hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \hat{\beta}_3 X_3 + \hat{\beta}_4 X_4 + \hat{\beta}_5 X_5 + \hat{\epsilon} \quad (4.2)$$

The two Messi variables are Played X_4 , and Played Away X_5 , this can be referred to as the Messi impact. The first one, played, see's whether or not he was present in the match that was played at home represented by either a 1 or 0 respectively, and the second one Played Away showcases whether or not he was playing away during that match. The second variable was only needed to help define Messi's performance in away, as the played variable delineates his performance at home. The reason for adding his presence, as opposed towards the number of actual goals he made in a

game is because the overall effect of him just being in the match can be interpreted, whereas simply seeing his contributions and subtracting it to see whether they would win or not, is not only rudimentary but unrealistic, as he would simply be replaced with a player of similar stature. With this core model now implemented, because of making use of a Poisson Regression, it was easily extendable to include an aggression factor to help take into account the number of fouls in order to determine how well Messi's performance was based on the aggressive nature of the match. This is found in the equation 4.3.

$$\log_e \hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \hat{\beta}_3 X_3 + \hat{\beta}_4 X_4 + \hat{\beta}_5 X_5 + \hat{\beta}_6 X_6 + \hat{\beta}_7 X_7 + \hat{\beta}_8 X_8 + \hat{\epsilon} \quad (4.3)$$

This particular equation is an extension of equation 4.2 and introduces three new variables, the aggressiveness of the match based on the number of fouls X_5 , the interaction between Messi's presence and the aggression X_6 , and the interaction between Messi playing away and the aggression X_7 . The final variable was simply added to help determine Messi's ability to score when he's not only playing away, but also when he's faced with the aggressive factor of a team. The reason for including the number of fouls in the match instead of the number of yellow or red cards was first yellow cards are simply controlled by the referee creating a referee factor that would also need to be taken account. Second red cards are rarely awarded during a season with an average of around 0.055 being awarded in a season, as seen in Figure 3.3. variable hr and ar. Fouls is a good measure, as any form of contact during a game that could have resulted in near injury of the player, helps define aggression in a more simpler and coherent way.

Once this model is implemented, the next model to be made use of is the ELO ratings system. The model was designed initially to help rank chess players, but was later modified to be adopted in the realm of sports. Each team by the end of the season will have an ELO rating where a higher value means a stronger team and

a lower value means a weaker team, thus giving a concrete comparison mechanic between teams of differing strength, as it takes into account several variables. After a match a team is assigned a new ELO rating based on their performance and previous ELO rating. From this model, the final ELO rating calculated for Barcelona is what is needed. The model consists of several variables which will very briefly be described. The core of the model can be seen in equation 4.4.

$$R_n = R_o + K * G(W - W_o) \quad (4.4)$$

The R_n variable is the new rating, the R_o variable is the old team's rating, the K is a constant value based on the League being played, the G variable is the number based on what the difference of goals is. If the difference is 0 or 1, G is 1, if the difference is by 2, the G is 1.5, if the difference is three or more goals, the G value is determined by $\frac{11+N}{8}$, where N is the number of goals. W variable is the result of the match, 1 for a win, 0.5 for a draw, and 0 for a loss. Lastly, W_e is the expected result of a match defined by an additional equation found in 4.5, where dr is the difference of both current ELO ratings of the teams playing, in where the home team's rating has an additional 100 points for strength in home. The main component that is being changed utilizing the Poisson Regression is the G , as the goals is what is being predicted for every game in a season where Messi was present and in the same season where his presence was removed. The predicted goals will be used to fill the FTHG and FTAG features of a data set for every single match played in that season and an ELO Ratings python script has been created to calculate the final ELOs of Barcelona in a season with Messi and a season without him.

$$W_e = \frac{1}{(10^{(-dr/400)} + 1)} \quad (4.5)$$

The final model is a simple linear regression model to help convert both these ELO

ratings towards La Liga ELO ratings adjusted for EPL Teams. This particular path was chosen because of the way the model is expandable, as given more time it can take into account several other features, but for the purpose of this paper a linear regression will suffice. The equation is found in 4.6.

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\epsilon} \quad (4.6)$$

The output Y is considered to be the La Liga ELO rating adjusted for when they play against EPL teams. The X_1 variable is the calculated ELO rating from the above models.

The reason for making use of three models that are interconnected with one another is that implementing an actual simulation of the Premier League is difficult and would require more time. Because of the robustness of ELO ratings as a ranking of how well teams were performing, a way to somehow integrate Lionel Messi was needed, therefore the Poisson Regression was devised. As the basic model was already made use to help determine the expected amount of goals, the flexibility of the model helped to add Messi as a factor and his response towards the aggressiveness to a game, thereby giving the total number of expected goals. Using these expected goals in calculating the ELO gave the rankings of Barcelona with Messi's contributions and without. The final piece was to find a competition where EPL teams play La Liga teams to see how La Liga teams fair against EPL teams. Using ELO ratings again for this would show what La Liga team ratings are based on when they play against EPL teams, thus allowing for a formula to adjust the ELO ratings calculated from the interaction of the other two models for when La Liga teams play EPL teams.

Chapter 5

Results

Once the Poisson Regression model was fitted, the resulting coefficients that were found can be seen in figure 5.1. From these coefficients, certain interpretations give an insight to how Lionel Messi contributes towards the overall number of expected goals during the season of 2016 to 2017.

Dep. Variable:	goals	No. Observations:	760
Model:	GLM	Df Residuals:	717
Model Family:	Poisson	Df Model:	42
Link Function:	log	Scale:	1.0000
Method:	IRLS	Log-Likelihood:	-1081.4
Date:	Fri, 19 Apr 2019	Deviance:	761.46
Time:	15:03:12	Pearson chi2:	646.

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.3418	0.254	-1.348	0.178	-0.839	0.155
home	0.2649	0.060	4.380	0.000	0.146	0.383
Played	0.6236	0.155	4.022	0.000	0.320	0.927
PlayedAway	0.0825	0.330	0.250	0.803	-0.565	0.730
Aggression	-0.0007	0.009	-0.076	0.940	-0.018	0.017
Aggression:Played	-0.0222	0.023	-0.957	0.339	-0.068	0.023
Aggression:PlayedAway	-0.0173	0.040	-0.431	0.666	-0.096	0.061

Figure 5.1: Poisson Regression Values

From the table the Played coefficient is 0.2649, where holding everything else constant conveys that when Messi plays at home, he contributes a total of $e^{0.2649}$ goals. The PlayedAway coefficient was found to be 0.0825, showcasing that Messi

plays away he is only able to contribute $e^{0.0825}$ goals, which is much less, thus sketching the fact that he plays better at home. From the Aggression factors, one can infer that for every foul that occurs it reduces Messi's ability to score at home by a factor of -0.0222 and when he plays away it reduces it by 0.0173. This helps convey the fact that Messi does not perform well when the aggression of a match increases and that the effect is far worse when he is playing at home. Even though the aggressive factor does penalize Messi for every foul, it is somewhat exaggerated as for every foul that occurs Messi expected goals reduce, and as there are nearly on average 14 fouls per match, from figure 3.2 and thus seems inaccurate that Messi would be so effected by one foul.

Running this fit to predict the number of expected goals when Messi is present in the season and when he is purposely removed gives two sets of expected goals. For all matches excluding Messi's team Barcelona the expected goals remains the same, but for matches concerning the player of this discussion, the goals vary to show Messi's impact. The differences between the expected goals of each match Messi plays versus the same matches where he is removed is shown in Figure 5.2. This goes to show that Messi being present helps the team score more goals, as the majority of matches where Messi is playing, represented by the blue bars, is usually higher in terms of number of goals scored than if he were not playing as represented by the orange bars.

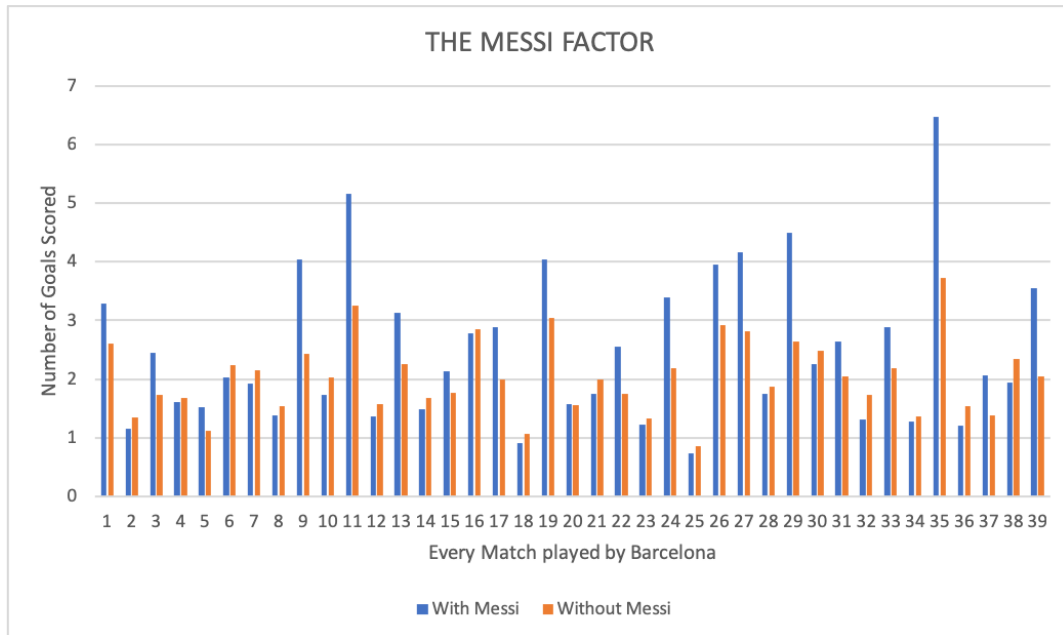


Figure 5.2: The Messi Impact

Using the estimated goals from the two separate data sets with one having the expected goals with Messi and the other without, the final ELO Ratings of Barcelona are reported to be 1610.978234 with Messi and 1538.912 without Messi. This helps bolster the fact that Messi does aid his team when he is present compared to when he is not. Figure 5.3 and 5.4 showcase a simulation of La Liga and the ELO scores of every team, where figure 5.3 is a season with Messi and 5.4 is the same season with Messi removed.

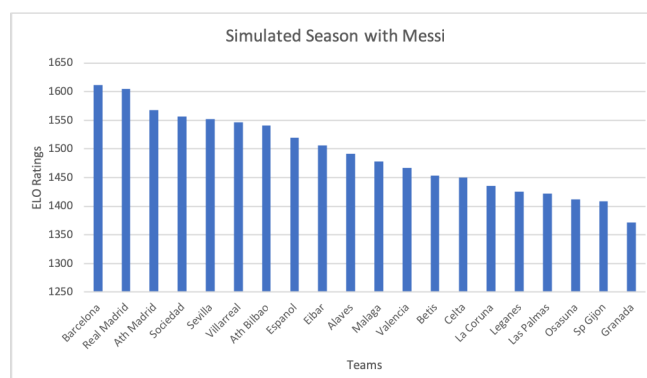


Figure 5.3: Season 2016-2017 with Messi

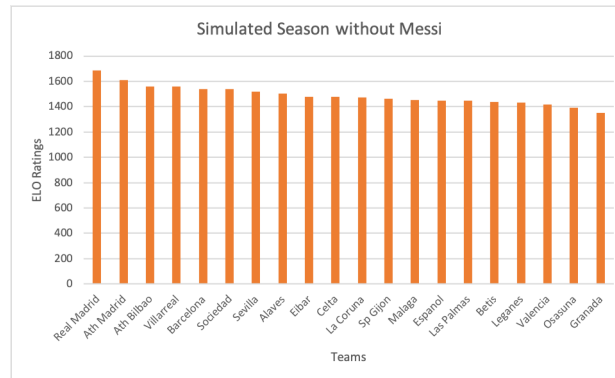


Figure 5.4: Season 2016-2017 without Messi

From each respective graph, one can see that when Messi was present, Barcelona was at the top of the season attaining an ELO rating of 1610.98, whereas without Messi, they were placed only fourth with a reported ELO Rating of 1538.91, far less than the previous. This helps convey the fact that Messi is a highly valued player within his own League, as without him, his home team does not even come within the top three teams of the league.

With these new ELO ratings acquired, the ELO rating with Messi present can be used to determine how he would fair against EPL teams once adjusted. Figure 5.5 draws the conversion between a La Liga ELO rating and an adjusted La Liga rating for the EPL, found through a simple linear regression model. This model was found by determining the ELO Rating of La Liga teams in the Champions League and using the La Liga ELO ratings of the season 2016 to 2017 and finding a basic coefficient that converts the latter to the previous.

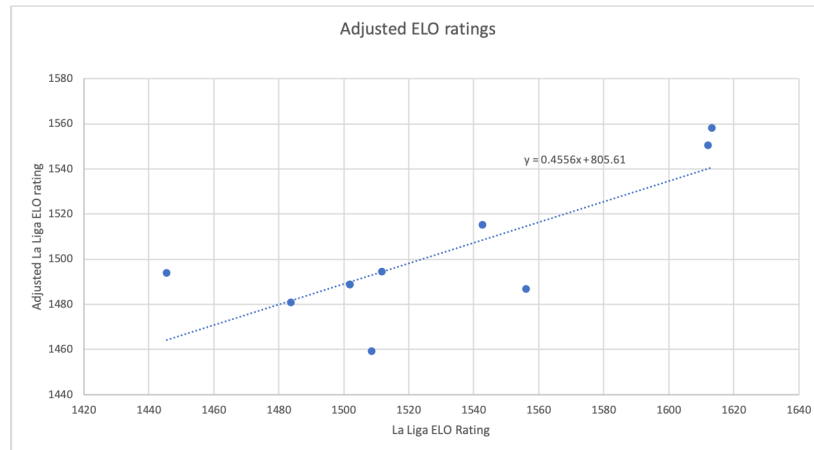


Figure 5.5: ELO Converter

Using the regression of $y = 0.4556x + 805.61$, the reported ELO values are 1539.572488 and 1506.737296, with the previous being the ELO rating with Messi and the later being without. There are certain problems with the linear regression as the simplicity of just comparing ELO ratings of La Liga teams in the Champions league and in their own league to formulate this conversion is simplistic in nature, and does not take into account several other factors, nonetheless it does give some idea of where Messi stands against EPL teams. Figure 5.6 demonstrates how Barcelona ranks against other teams in the EPL with Messi and figure 5.7 without.

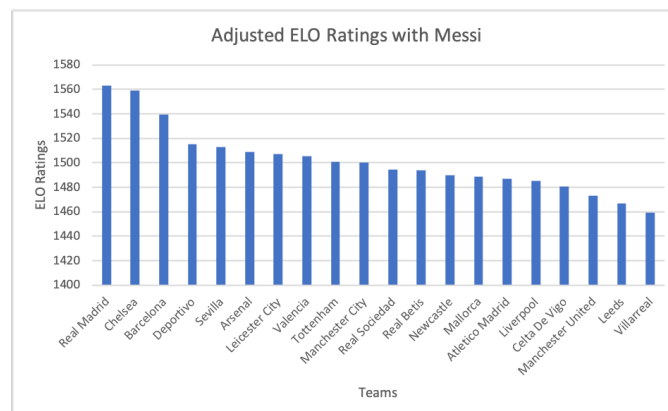


Figure 5.6: Barcelona versus. EPL Teams with Messi

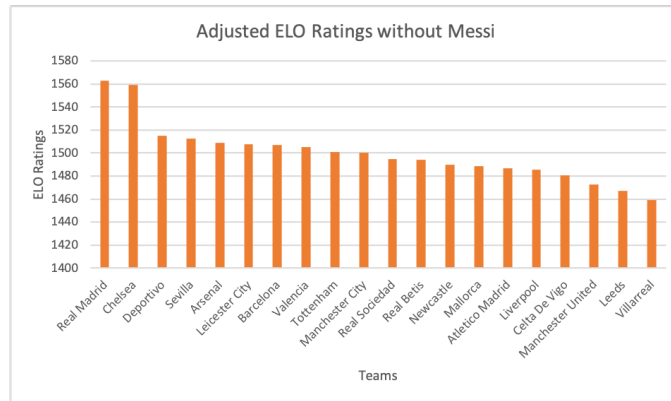


Figure 5.7: Barcelona versus. EPL Teams without Messi

In figure 5.6, Barcelona ranks top of all EPL teams except for Chelsea when Messi plays with them putting them at a formidable position against other EPL teams, with the next strongest EPL team, Arsenal only attaining an ELO rating of 1508.67 nearly 20 points behind Barcelona. However, when Messi is absent, Barcelona ranks much further down with teams that were considered ranking lower in Figure 5.6 being ranked much higher. Barcelona's adjusted ELO rating places them near the middle of the graph leaving them at a much more average status as opposed towards when Messi is playing, as EPL teams between Barcelona and Liverpool, such as Tottenham, Manchester City. and Newcastle, are not ranked too far off from Barcelona.

These results help answer the question of would Messi be considered a valuable player if he were to play in the EPL instead of La Liga, as conceived by the notion that the EPL ranking of Barcelona adjusted towards a tournament of EPL and La liga teams fairs much higher when Messi is present as opposed towards when he was not. This is not only shown through the substantial difference between the ELO ratings but also how they rank against other teams, as Barcelona with Messi is considered a top tier team when playing against EPL teams, but without Messi simply blends with teams of an average rating, thus showcasing not only his immense impact, but also his ability to play well against EPL teams. All tables used to generate the figures are in the Appendix A located in the final page of this report.

Chapter 6

Discussion

The results attained from the previous section determine that Lionel Messi is considered a valuable player even if he plays in the EPL. This is seen through how his team's ELO rating fairs against EPL teams with and without him present. However, while the results may seem compelling there are several ways to improve certain aspects of the model to make it more robust. Before delving into how improvements can be made, a detailed explanation of the results will first be explored.

The result that was found through the three step analysis was that the ELO rating of Barcelona adjusted towards a competition, the Champions League, where La Liga teams interact with EPL teams, was calculated to be 1539.57 with Messi and 1506.73 without. What these results mean is that when Messi is playing for Barcelona, the team ranks much higher against Premier League teams than if the team was not playing for Barcelona. This resulting difference can help show how crucial Messi is towards his own team and that it's due to him that the adjusted ELO of his team is top tier, whereas without him the team is slightly above average. The difference in adjusted ELO Ratings is nearly 33 points, thus delineating the fact that Messi contributed to around 33 points of ELO, which in the context of ELO ratings can differentiate a team by several ranks as witnessed through Figure 5.6 and 5.7, in the previous section, in where Barcelona ranking falls from third to seventh, a drop of 4 ranks with a comparison to 20 other teams.

There are number of issues with the model that need to be rectified and explored in order to make it more accurate and robust. The first problem is the model simply looks at whether or not Messi was present during a game. The main problem with this is if other players were also coincidentally present during every single match that Messi was the model could potentially apply to them. In order to help factor this out other features need to be incorporated into this model, some potential variables could be the number of passes, the number of successful versus failed shots he takes etc...Additionally to help include more variables to determine the level of difficulty towards a match a feature selection algorithm could be made use of to determine strong correlations between expected goals and certain features in the data set.

To further add, another step of the model that also holds various problems is the ELO converter, the linear regression that adjusts La Liga ELO ratings towards EPL teams. The first problem is the data. While the Champions league does illustrate the interaction between EPL teams and La Liga teams, the competition is usually only played by the four best teams from each league, therefore the number of matches being played is lacking and because of that the model is less trained. Because of the volatile nature of the EPL teams that reach the top four, certain teams that were in the Champions league may not be there, thus leading to a lack of games played for some teams thus inhibiting their ELO scores from changing. The second problem is the lack of features utilized, to better help make it more robust more features need to be integrated such as an aggression factor and for that more data essentially needs to be acquired and formulated.

Chapter 7

Conclusion

In conclusion, from the resulting model generated to help determine an answer, Messi would still in fact be considered a valuable player if he played in the English Premier League instead of La Liga. This is evident in the substantial difference in the adjusted ELO ratings calculated for when Messi was present versus. from when he is not. The resulting ELO ratings were found to be 1539.572488 and 1506.737296 respectively, and this results in a ranking difference between Barcelona with Messi and Barcelona without Messi against other EPL teams. Nonetheless, while the analysis does help answer the question there are several ways to make the model more robust and expand it to make it more accurate as discussed in the previous chapter.

Chapter 8

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Appendix A

Tables

La Liga ELO Ratings with Messi	
Barcelona	1610.98
Real Madrid	1604.89
Ath Madrid	1567.29
Sociedad	1556.24
Sevilla	1551.63
Villarreal	1546.44
Ath Bilbao	1541.07
Espanol	1519.98
Eibar	1506.23
Alaves	1491.37
Malaga	1478.06
Valencia	1466.95
Betis	1453.93
Celta	1450.06
La Coruna	1435.44
Leganes	1425.01
Las Palmas	1422.38
Osasuna	1412.45
Sp Gijon	1408.03
Granada	1371.08

Figure A.1: Table corresponding to Figure 5.3

La Liga Ratings without Lionel Messi	
Real Madrid	1687.54
Ath Madrid	1610.81
Ath Bilbao	1558.78
Villarreal	1557.03
Barcelona	1538.91
Sociedad	1537.71
Sevilla	1520.05
Alaves	1502.88
Eibar	1479.83
Celta	1478.9
La Coruna	1475.2
Sp Gijon	1460.16
Malaga	1454.71
Espanol	1448.45
Las Palmas	1447.35
Betis	1439.02
Leganes	1434
Valencia	1416.58
Osasuna	1392.49
Granada	1351.26

Figure A.2: Table corresponding to Figure 5.4

Barcelona's adjusted ELO ratings compared to EPL teams with Messi		
Real Madrid	1562.85	
Chelsea	1559.34	
Barcelona	1539.57	
Deportivo	1515.17	
Sevilla	1512.82	
Arsenal	1508.67	
Leicester City	1507.36	
Valencia	1505.26	
Tottenham	1500.77	
Manchester City	1500.44	
Real Sociedad	1494.54	
Real Betis	1494.03	
Newcastle	1489.59	
Mallorca	1488.73	
Atletico Madrid	1486.75	
Liverpool	1485.33	
Celta De Vigo	1480.75	
Manchester United	1472.83	
Leeds	1467.01	
Villarreal	1459.24	

Figure A.3: Table corresponding to Figure 5.6

Barcelona's adjusted ELO ratings compared to EPL teams without Messi			
Real Madrid	1562.9		
Chelsea	1559.3		
Deportivo	1515.2		
Sevilla	1512.8		
Arsenal	1508.7		
Leicester City	1507.4		
Barcelona	1506.7		
Valencia	1505.3		
Tottenham	1500.8		
Manchester City	1500.4		
Real Sociedad	1494.5		
Real Betis	1494		
Newcastle	1489.6		
Mallorca	1488.7		
Atletico Madrid	1486.8		
Liverpool	1485.3		
Celta De Vigo	1480.8		
Manchester United	1472.8		
Leeds	1467		
Villarreal	1459.2		

Figure A.4: Table corresponding to Figure 5.7